

Can 50 Logical Qubits Run Serious Quantum-Dot Benchmarks on Magne?

VQE–UCCSD initial states feeding the Rodeo algorithm

Feasibility assessment for M. Hjorth-Jensen & the EIGEN-Q team

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N electrons in a 2D harmonic trap + Coulomb · HO basis of K spin-orbitals · Jordan–Wigner

Bottom line up front

The one-sentence answer

For 2D quantum dots, **qubit count is not the bottleneck** — 50 logical qubits comfortably hold the largest natural basis ($K = 42$ spin-orbitals, shells 0–5) for $N = 2, 6, 12$ electrons, with ancillas to spare. The limiter is **circuit depth**, and these runs will *validate* the pipeline rather than beat classical methods.

- Mapping here is **1 qubit per spin-orbital** (not 2/site as in Hubbard): K spin-orbitals $\rightarrow K$ qubits.
- **~12 qubits** already gives a non-trivial, exactly-checkable correlation benchmark (the notebook's $K = 12$ case).
- **~42 qubits** reaches the full shells-0–5 space — a $\sim 10^{10}$ -determinant FCI, right at the classical exact-FCI frontier.
- No quantum advantage: coupled cluster / DMRG / selected-CI own this regime. Value = **clean pipeline validation** and high-overlap state prep for Rodeo.

The encoding: K spin-orbitals $\rightarrow K$ qubits

Each harmonic-oscillator shell $N_s = 2n + |m|$ holds $2(N_s+1)$ spin-orbitals. Under Jordan–Wigner, **one spin-orbital = one qubit**.

shells	spin-orb. K	qubits	magic N
0	2	2	2
0–1	6	6	6
0–2	12	12	12
0–3	20	20	20
0–4	30	30	30
0–5	42	42	42
0–6	56	56	56

Headroom on Magne

50 logical qubits $\Rightarrow$ active space up to $K \approx 47\text{--}49$ spin-orbitals (after 1–few Rodeo ancillas).

- Shells 0–5 ($K = 42$): **fits comfortably**.
- Shells 0–6 ($K = 56$): **does not fit**.

So the natural maximal dot active space on Magne is $K = 42$.

Qubit budget vs. classical cost

N	qubits K	shells	FCI dim $\binom{K}{N}$	classical status
2	12	0–2	66	trivial
2	20	0–3	190	trivial
2	42	0–5	861	trivial
6	20	0–3	38,760	trivial
6	30	0–4	5.9×10^5	trivial
6	42	0–5	5.2×10^6	easy exact FCI
12	30	0–4	8.6×10^7	easy exact FCI
12	42	0–5	1.1×10^{10}	exact-FCI frontier
20	42	0–5	5.1×10^{11}	beyond exact FCI (CC/DMRG ok)

Symmetry sectors (M, S_z) shrink these $\sim 5\text{--}20\times$. Note: at the magic numbers N sits far from half-filling, so FCI dimensions stay modest — the dot is *not* a worst-case correlation problem.

Where the classical frontier is for quantum dots

- Exact FCI is feasible to $\binom{K}{N} \sim 10^{10}$ determinants — i.e. up to the $N=12$, $K=42$ case *on a workstation/cluster*.
- Beyond that, **coupled cluster (CCSD(T))**, DMRG, selected-CI and AFQMC are accurate *and cheap* for these systems.
- Quantum dots in this notebook's regime are **smooth and moderately correlated**: HF is a good reference, correlation converges by $K \approx 12$, UCCSD recovers it in a ~ 7 -parameter landscape.

Consequence

There is **no quantum-dot regime reachable with 50 logical qubits that beats classical methods**. This is expected — and fine. The scientific role of these runs is **algorithm validation against an exact reference**, and demonstrating high-overlap initial states for Rodeo.

So how many logical qubits for a “serious” benchmark?

Benchmark tier	qubits	What it demonstrates
Sanity / shakedown	2–6	JW Hamiltonian, Z-basis readout, HF state prep.
Meaningful	12	First real correlation (notebook’s $K=12$, ~ 7 UCCSD amplitudes); exact FCI check; full VQE→Rodeo loop.
Substantial	20–30	$N=2, 6$ in larger bases; non-trivial UCCSD depth; approaches basis-set convergence.
Maximal natural	42	Shells 0–5; $N=12$ is a $\sim 10^{10}$ -det FCI — a genuinely meaty benchmark, <i>if</i> depth allows.

Verdict on the qubit axis

12 qubits for a credible first benchmark, 20–30 for a substantial one, 42 for the largest natural dot space — all inside 50. Qubit count is not the limiter.

The binding constraint is depth (again)

Raw Trotterised UCCSD two-qubit-gate counts under Jordan–Wigner (1 Trotter step):

N	K	singles	doubles	~ 2 -qubit gates
2	12	20	45	$\sim 10^4$
6	20	84	1,365	$\sim 4 \times 10^5$
12	30	216	10,098	$\sim 5 \times 10^6$
12	42	360	28,710	$\sim 2 \times 10^7$

- JW adds non-local Z -strings of length up to K per term — neutral-atom reconfigurable connectivity helps, but depth still grows fast.
- Rodeo adds M rounds of controlled- e^{-iHt_j} ; H has $\sim 10^2$ ($K=12$) to $\sim 10^5$ ($K=42$) Pauli terms per Trotter step.
- **Depth-feasible on an early Level-2 machine: $K=12$, perhaps $K=20$ with ADAPT-VQE.** $K=30$ – 42 are qubit-feasible but *depth-prohibitive* at first.

Recommendations

Make depth, not width, the design driver

- Use **ADAPT-VQE** / symmetry-adapted pools: the raw 28,710 doubles at $K=42$ collapse to $O(10-100)$ selected operators.
- Exploit the **division of labour**: VQE need only reach overlap $p \approx 0.9-0.95$ — *Rodeo* supplies the precision. A *shallow* UCCSD that gets high overlap is enough; do not over-optimize VQE depth.

Choose friendly physics & honest claims

- Work at **moderate** $\hbar\omega$ (weak/intermediate coupling) where HF is a good reference; avoid the small- $\hbar\omega$ Wigner-molecule regime that needs deeper circuits.
- In the application, replace “20–40 qubits” with explicit (N, K) : e.g. “ $N=2-12$ dots in active spaces up to $K=42$ spin-orbitals (≤ 42 qubits).”
- Always pair a run with its **exact-FCI reference** — the dot’s great virtue.

A concrete benchmark ladder for Magne

- 1 $K = 2$ ($N = 2$) — hardware sanity: HF state, diagonal H , Rodeo on a 1-state space.
- 2 $K = 6$ ($N = 2$) — first off-diagonal scattering; validate VQE energy vs. FCI (dim 15).
- 3 $K = 12$ ($N = 2$) — **the flagship first benchmark.** ~ 7 UCCSD amplitudes, FCI dim 66, full VQE $\rightarrow$ Rodeo with exact reference. ~ 12 *logical qubits*.
- 4 $K = 20$ ($N = 2$ or 6) — ADAPT-VQE; measure rodeo convergence vs. rounds M . ~ 20 *qubits*.
- 5 **Stretch:** $K = 30\text{--}42$ ($N = 6, 12$) — only as the logical depth budget allows; $N=12, K=42$ is the $\sim 10^{10}$ -det showcase. $30\text{--}42$ *qubits*.

Every rung has an exact classical answer, so each one cleanly separates *algorithm error* from *hardware error* — exactly what a Level-2 demonstration needs.

Closing summary

Feasibility of quantum dots on Magne (50 logical qubits)

Qubit-feasible across the board: 12 qubits for a meaningful benchmark, 20–30 for a substantial one, 42 for the full shells-0–5 space — all within budget. **Depth-limited:** only the smaller cases ($K \leq 20$, with ADAPT-VQE) will run to convergence on an early Level-2 machine. **No classical advantage** — and none expected.

- The quantum dot is the **ideal validation system**: exact FCI reference at every size, smooth correlation, and a clean test of VQE–UCCSD → Rodeo end-to-end.
- Spend the qubits you have on **depth-reducing structure** (ADAPT, symmetry, shallow high-overlap VQE), not on the largest possible basis.
- Use it to **characterise Magne and de-risk the pipeline** before the harder Hubbard / nuclear targets.